NOTRE DAME UNIVERSITY BANGLADESH

Machine Learning Lab Report-01

Course Code: CSE4214

Course Title: Machine Learning Lab

Lab Task Topic: Data Loading and Preprocessing Using Pandas

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Objective

The objective of this experiment is to familiarize with basic data handling and preprocessing techniques required before applying Machine Learning algorithms. This lab focuses on loading a dataset, inspecting its structure, handling categorical variables, and converting them into numerical representations suitable for model training.

Dataset Description

The dataset used in this experiment is `bank.csv`, which contains customer information related to a banking institution. The dataset includes demographic details, financial attributes, and campaign-related information. The target variable indicates whether a customer subscribed to a term deposit. The main objective is to load a CSV dataset into a Pandas DataFrame and inspect the initial records.

1 Loading CSV Data

[1]:
```
import pandas as pd

df = pd.read_csv("bank.csv")
df.head()
```

[1]:
```
  age;"job";"marital";"education";"default";"balance";"housing";"loan";
 ↪"contact"
;"day";"month";"duration";"campaign";"pdays";"previous";"poutcome";"y"
0  30;"unemployed";"married";"primary";"no";1787;…
1  33;"services";"married";"secondary";"no";4789;…
2  35;"management";"single";"tertiary";"no";1350;…
3  30;"management";"married";"tertiary";"no";1476…
4  59;"blue-collar";"married";"secondary";"no";0;…
```

Explanation

Pandas provides the `read_csv()` function to load structured data into a DataFrame. Initially, the dataset was loaded without specifying the delimiter, which resulted in improper column parsing. This step helps identify formatting issues in raw data.

Output

The output displays the first five records of the dataset, revealing incorrect column separation due to the delimiter mismatch.

[2]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
---------------------------------------------------------------------------
ModuleNotFoundError                       Traceback (most recent call last)
Cell In[2], line 4
      2 import pandas as pd
```

```
      3 import matplotlib.pyplot as plt
----> 4 import seaborn as sns

ModuleNotFoundError: No module named 'seaborn'
```

[3]:
```
df = pd.read_csv("bank.csv", sep = ';')
df.head()
```

[3]:
```
   age           job  marital  education default  balance housing loan  \
0   30    unemployed  married    primary      no     1787      no   no
1   33      services  married  secondary      no     4789     yes  yes
2   35    management   single   tertiary      no     1350     yes   no
3   30    management  married   tertiary      no     1476     yes  yes
4   59   blue-collar  married  secondary      no        0     yes   no

    contact  day month  duration  campaign  pdays  previous poutcome   y
0  cellular   19   oct        79         1     -1         0  unknown  no
1  cellular   11   may       220         1    339         4  failure  no
2  cellular   16   apr       185         1    330         1  failure  no
3   unknown    3   jun       199         4     -1         0  unknown  no
4   unknown    5   may       226         1     -1         0  unknown  no
```

[4]:
```
df.tail()
```

[4]:
```
      age            job  marital  education default  balance housing loan  \
4516   33       services  married  secondary      no     -333     yes   no
4517   57  self-employed  married   tertiary     yes    -3313     yes  yes
4518   57     technician  married  secondary      no      295      no   no
4519   28    blue-collar  married  secondary      no     1137      no   no
4520   44   entrepreneur   single   tertiary      no     1136     yes  yes

       contact  day month  duration  campaign  pdays  previous poutcome   y
4516  cellular   30   jul       329         5     -1         0  unknown  no
4517   unknown    9   may       153         1     -1         0  unknown  no
4518  cellular   19   aug       151        11     -1         0  unknown  no
4519  cellular    6   feb       129         4    211         3    other  no
4520  cellular    3   apr       345         2    249         7    other  no
```

[5]:
```
def replace_marital(val):
  if val == 'single':
    return 0
  else:
    return 1

df['marital'] = df['marital'].apply(replace_marital)
df.head()
```

[5]:
```
   age           job  marital  education default  balance housing loan  \
0   30    unemployed        1    primary      no     1787      no   no
1   33      services        1  secondary      no     4789     yes  yes
2   35    management        0   tertiary      no     1350     yes   no
```

```
3   30   management        1   tertiary      no     1476     yes  yes
4   59  blue-collar        1  secondary      no        0     yes   no

    contact  day month  duration  campaign  pdays  previous poutcome   y
0  cellular   19   oct        79         1     -1         0  unknown  no
1  cellular   11   may       220         1    339         4  failure  no
2  cellular   16   apr       185         1    330         1  failure  no
3   unknown    3   jun       199         4     -1         0  unknown  no
4   unknown    5   may       226         1     -1         0  unknown  no
```

[6]:
```
df["housing"] = df["housing"].map({"yes": 1, "no": 0}.get)
df.head()
```

[6]:
```
   age          job  marital  education default  balance  housing loan  \
0   30   unemployed        1    primary      no     1787        0   no
1   33     services        1  secondary      no     4789        1  yes
2   35   management        0   tertiary      no     1350        1   no
3   30   management        1   tertiary      no     1476        1  yes
4   59  blue-collar        1  secondary      no        0        1   no

    contact  day month  duration  campaign  pdays  previous poutcome   y
0  cellular   19   oct        79         1     -1         0  unknown  no
1  cellular   11   may       220         1    339         4  failure  no
2  cellular   16   apr       185         1    330         1  failure  no
3   unknown    3   jun       199         4     -1         0  unknown  no
4   unknown    5   may       226         1     -1         0  unknown  no
```

[7]:
```
df["loan"] = df["loan"].replace({"yes": 1, "no": 0})
df.head()
```

```
/tmp/ipykernel_18086/1568368702.py:1: FutureWarning: Downcasting behavior in
`replace` is deprecated and will be removed in a future version. To retain the
old behavior, explicitly call `result.infer_objects(copy=False)`. To opt-in to
the future behavior, set `pd.set_option('future.no_silent_downcasting', True)`
  df["loan"] = df["loan"].replace({"yes": 1, "no": 0})
```

[7]:
```
   age          job  marital  education default  balance  housing  loan  \
0   30   unemployed        1    primary      no     1787        0     0
1   33     services        1  secondary      no     4789        1     1
2   35   management        0   tertiary      no     1350        1     0
3   30   management        1   tertiary      no     1476        1     1
4   59  blue-collar        1  secondary      no        0        1     0

    contact  day month  duration  campaign  pdays  previous poutcome   y
0  cellular   19   oct        79         1     -1         0  unknown  no
1  cellular   11   may       220         1    339         4  failure  no
2  cellular   16   apr       185         1    330         1  failure  no
3   unknown    3   jun       199         4     -1         0  unknown  no
4   unknown    5   may       226         1     -1         0  unknown  no
```

[22]:
```
df["loan"] = df["loan"].replace({
"yes": 1,
"no": 0})
df.head()
```

[22]:
```
   age  job  marital  education default  balance  housing  loan   contact  \
0   30  0.0        1        1.0      no     1787        0     0  cellular
1   33  0.0        1        2.0      no     4789        1     1  cellular
2   35  1.0        0        3.0      no     1350        1     0  cellular
3   30  1.0        1        3.0      no     1476        1     1   unknown
4   59  0.0        1        2.0      no        0        1     0   unknown

   day  month  duration  campaign  pdays  previous poutcome   y
0   19    NaN        79         1     -1         0  unknown  no
1   11    NaN       220         1    339         4  failure  no
2   16    NaN       185         1    330         1  failure  no
3    3    NaN       199         4     -1         0  unknown  no
4    5    NaN       226         1     -1         0  unknown  no
```

[9]:
```
df["job"].unique()
```

[9]:
```
array(['unemployed', 'services', 'management', 'blue-collar',
       'self-employed', 'technician', 'entrepreneur', 'admin.', 'student',
       'housemaid', 'retired', 'unknown'], dtype=object)
```

[18]:
```
df["job"] = df["job"].replace({'unemployed': 0,
                    'services': 0,
                    'management': 1,
                    'blue-collar': 0,
                    'self-employed': 0,
                    'technician': 1,
                    'entrepreneur': 1,
                    'admin.': 0,
                    'student': 1,
                    'housemaid': 0,
                    'retired': 0,
                    'unknown': np.nan})
df.head()
```

[18]:
```
   age  job  marital  education default  balance  housing  loan   contact  \
0   30  0.0        1        1.0      no     1787        0     0  cellular
1   33  0.0        1        2.0      no     4789        1     1  cellular
2   35  1.0        0        3.0      no     1350        1     0  cellular
3   30  1.0        1        3.0      no     1476        1     1   unknown
4   59  0.0        1        2.0      no        0        1     0   unknown

   day  month  duration  campaign  pdays  previous poutcome   y
0   19     10        79         1     -1         0  unknown  no
1   11      5       220         1    339         4  failure  no
2   16      4       185         1    330         1  failure  no
3    3      6       199         4     -1         0  unknown  no
```

```
4    5       5        226          1      -1          0  unknown  no
```

[11]:
```
df["month"].unique()
```

[11]:
```
array(['oct', 'may', 'apr', 'jun', 'feb', 'aug', 'jan', 'jul', 'nov',
       'sep', 'mar', 'dec'], dtype=object)
```

[20]:
```
df.month = df.month.map({
    'oct': 10,
    'may': 5,
    'apr': 4,
    'jun': 6,
    'feb': 2,
    'aug': 8,
    'jan': 1,
    'jul': 7,
    'nov': 11,
    'sep': 9,
    'mar': 3,
    'dec': 12
})
df.head(10)
```

[20]:
```
   age  job  marital  education default  balance  housing  loan   contact  \
0   30  0.0        1        1.0      no     1787        0     0  cellular
1   33  0.0        1        2.0      no     4789        1     1  cellular
2   35  1.0        0        3.0      no     1350        1     0  cellular
3   30  1.0        1        3.0      no     1476        1     1   unknown
4   59  0.0        1        2.0      no        0        1     0   unknown
5   35  1.0        0        3.0      no      747        0     0  cellular
6   36  0.0        1        3.0      no      307        1     0  cellular
7   39  1.0        1        2.0      no      147        1     0  cellular
8   41  1.0        1        3.0      no      221        1     0   unknown
9   43  0.0        1        1.0      no      -88        1     1  cellular

   day  month  duration  campaign  pdays  previous poutcome   y
0   19    NaN        79         1     -1         0  unknown  no
1   11    NaN       220         1    339         4  failure  no
2   16    NaN       185         1    330         1  failure  no
3    3    NaN       199         4     -1         0  unknown  no
4    5    NaN       226         1     -1         0  unknown  no
5   23    NaN       141         2    176         3  failure  no
6   14    NaN       341         1    330         2    other  no
7    6    NaN       151         2     -1         0  unknown  no
8   14    NaN        57         2     -1         0  unknown  no
9   17    NaN       313         1    147         2  failure  no
```

[13]:
```
df["education"].unique()
```

[13]:
```
array(['primary', 'secondary', 'tertiary', 'unknown'], dtype=object)
```

[14]:
```
df.education = df.education.map({
    'primary': 1,
    'secondary': 2,
    'tertiary': 3,
    'unknown': np.nan
})
```

[19]:
```
df.head(10)
```

[19]:
```
   age  job  marital  education default  balance  housing  loan   contact  \
0   30  0.0        1        1.0      no     1787        0     0  cellular
1   33  0.0        1        2.0      no     4789        1     1  cellular
2   35  1.0        0        3.0      no     1350        1     0  cellular
3   30  1.0        1        3.0      no     1476        1     1   unknown
4   59  0.0        1        2.0      no        0        1     0   unknown
5   35  1.0        0        3.0      no      747        0     0  cellular
6   36  0.0        1        3.0      no      307        1     0  cellular
7   39  1.0        1        2.0      no      147        1     0  cellular
8   41  1.0        1        3.0      no      221        1     0   unknown
9   43  0.0        1        1.0      no      -88        1     1  cellular

   day  month  duration  campaign  pdays  previous poutcome   y
0   19     10        79         1     -1         0  unknown  no
1   11      5       220         1    339         4  failure  no
2   16      4       185         1    330         1  failure  no
3    3      6       199         4     -1         0  unknown  no
4    5      5       226         1     -1         0  unknown  no
5   23      2       141         2    176         3  failure  no
6   14      5       341         1    330         2    other  no
7    6      5       151         2     -1         0  unknown  no
8   14      5        57         2     -1         0  unknown  no
9   17      4       313         1    147         2  failure  no
```

[23]:
```
df["poutcome"].unique()
```

[23]:
```
array(['unknown', 'failure', 'other', 'success'], dtype=object)
```

[24]:
```
df.poutcome = df.poutcome.map({
    'unknown': np.nan,
    'failure': 1,
    'other': 2,
    'success': 3
})
df.head(10)
```

[24]:
```
   age  job  marital  education default  balance  housing  loan   contact  \
0   30  0.0        1        1.0      no     1787        0     0  cellular
1   33  0.0        1        2.0      no     4789        1     1  cellular
2   35  1.0        0        3.0      no     1350        1     0  cellular
3   30  1.0        1        3.0      no     1476        1     1   unknown
4   59  0.0        1        2.0      no        0        1     0   unknown
```

```
5   35  1.0        0        3.0      no      747        0     0  cellular
6   36  0.0        1        3.0      no      307        1     0  cellular
7   39  1.0        1        2.0      no      147        1     0  cellular
8   41  1.0        1        3.0      no      221        1     0   unknown
9   43  0.0        1        1.0      no      -88        1     1  cellular

   day  month  duration  campaign  pdays  previous  poutcome   y
0   19    NaN        79         1     -1         0       NaN  no
1   11    NaN       220         1    339         4       1.0  no
2   16    NaN       185         1    330         1       1.0  no
3    3    NaN       199         4     -1         0       NaN  no
4    5    NaN       226         1     -1         0       NaN  no
5   23    NaN       141         2    176         3       1.0  no
6   14    NaN       341         1    330         2       2.0  no
7    6    NaN       151         2     -1         0       NaN  no
8   14    NaN        57         2     -1         0       NaN  no
9   17    NaN       313         1    147         2       1.0  no
```

[25]:
```
df["balance"] = df["balance"].apply(lambda v: (v - df["balance"].min()) /
 ↪(df["balance"].max() - df["balance"].min()))
```

[26]:
```
df.head(10)
```

[26]:
```
   age  job  marital  education default   balance  housing  loan   contact  \
0   30  0.0        1        1.0      no  0.068455        0     0  cellular
1   33  0.0        1        2.0      no  0.108750        1     1  cellular
2   35  1.0        0        3.0      no  0.062590        1     0  cellular
3   30  1.0        1        3.0      no  0.064281        1     1   unknown
4   59  0.0        1        2.0      no  0.044469        1     0   unknown
5   35  1.0        0        3.0      no  0.054496        0     0  cellular
6   36  0.0        1        3.0      no  0.048590        1     0  cellular
7   39  1.0        1        2.0      no  0.046442        1     0  cellular
8   41  1.0        1        3.0      no  0.047436        1     0   unknown
9   43  0.0        1        1.0      no  0.043288        1     1  cellular

   day  month  duration  campaign  pdays  previous  poutcome   y
0   19    NaN        79         1     -1         0       NaN  no
1   11    NaN       220         1    339         4       1.0  no
2   16    NaN       185         1    330         1       1.0  no
3    3    NaN       199         4     -1         0       NaN  no
4    5    NaN       226         1     -1         0       NaN  no
5   23    NaN       141         2    176         3       1.0  no
6   14    NaN       341         1    330         2       2.0  no
7    6    NaN       151         2     -1         0       NaN  no
8   14    NaN        57         2     -1         0       NaN  no
9   17    NaN       313         1    147         2       1.0  no
```

[27]:
```
df["pdays"] = df["pdays"].apply(lambda v: (v - df["pdays"].min()) /
 ↪(df["pdays"].max() - df["pdays"].min()))
```

[28]:
```
df.head(10)
```

[28]:

```
   age  job  marital  education default   balance  housing  loan   contact  \
0   30  0.0        1        1.0      no  0.068455        0     0  cellular
1   33  0.0        1        2.0      no  0.108750        1     1  cellular
2   35  1.0        0        3.0      no  0.062590        1     0  cellular
3   30  1.0        1        3.0      no  0.064281        1     1   unknown
4   59  0.0        1        2.0      no  0.044469        1     0   unknown
5   35  1.0        0        3.0      no  0.054496        0     0  cellular
6   36  0.0        1        3.0      no  0.048590        1     0  cellular
7   39  1.0        1        2.0      no  0.046442        1     0  cellular
8   41  1.0        1        3.0      no  0.047436        1     0   unknown
9   43  0.0        1        1.0      no  0.043288        1     1  cellular

   day  month  duration  campaign     pdays  previous  poutcome   y
0   19    NaN        79         1  0.000000         0       NaN  no
1   11    NaN       220         1  0.389908         4       1.0  no
2   16    NaN       185         1  0.379587         1       1.0  no
3    3    NaN       199         4  0.000000         0       NaN  no
4    5    NaN       226         1  0.000000         0       NaN  no
5   23    NaN       141         2  0.202982         3       1.0  no
6   14    NaN       341         1  0.379587         2       2.0  no
7    6    NaN       151         2  0.000000         0       NaN  no
8   14    NaN        57         2  0.000000         0       NaN  no
9   17    NaN       313         1  0.169725         2       1.0  no
```

[29]:

```
from sklearn.preprocessing import MinMaxScaler
```

[30]:

```
scaler = MinMaxScaler()
df["duration"] = scaler.fit_transform(df[["duration"]])
df["pdays"] = scaler.fit_transform(df[["pdays"]])
df.head()
```

[30]:

```
   age  job  marital  education default   balance  housing  loan   contact  \
0   30  0.0        1        1.0      no  0.068455        0     0  cellular
1   33  0.0        1        2.0      no  0.108750        1     1  cellular
2   35  1.0        0        3.0      no  0.062590        1     0  cellular
3   30  1.0        1        3.0      no  0.064281        1     1   unknown
4   59  0.0        1        2.0      no  0.044469        1     0   unknown

   day  month  duration  campaign     pdays  previous  poutcome   y
0   19    NaN  0.024826         1  0.000000         0       NaN  no
1   11    NaN  0.071500         1  0.389908         4       1.0  no
2   16    NaN  0.059914         1  0.379587         1       1.0  no
3    3    NaN  0.064548         4  0.000000         0       NaN  no
4    5    NaN  0.073486         1  0.000000         0       NaN  no
```

[ ]:

```

```